

Sidenor

MILL TEST CERTIFICATE

ISO 9001; IATF 16949; ISO 14001; ISO 45001; ISO 50001; ISO 27001 Y TISAX

Azkoitia Plant

SIDENOR ACEROS ESPECIALES S.L.U.
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Product Made in Spain

CUSTOMER: MAGELLAN CORPORATION	SALES ORDER: 352828-2	WORKS REFERENCE: 2259220
REFERENCE: 215567	MASTER REFERENCE: 294845	HEAT NUMBER: 643615
PRODUCT NR:		ROLLED: 30.03.2023

REQUIRED PRODUCT

NITRIDING 135 MOD ROUND BARS TURNED Q-T-STRESS RELIEVED 1.5030"

ASTM A-108 (") alloy 0.28% C=0.55 (U.R.N) 12"/12" NORMAL

EXPEDITION DELIVERY: 0080797399 WEIGHT (KG): 8.040 BUNDLES: 4 UNITS: 240

MADE ACCORDING TO

ASTM A108 18 01.11.2018; ASTM A255 10 2014; ASTM A29-A29M -20 01.07.2020; ASTM A370 -21 01.11.2021

ASTM A751 21 01.11.2021; ASTM E112 13 01.10.2013; ASTM E381 17 01.06.2017

CASTLE METALS N13500-01 17 01.06.2020; EN 10204 - OCT. 2004 3.1; MAGELLAN MIC2016 1 30.01.2012

MIL S6709A-NOT2 - 12.12.1994; SAE-AMS 2301 L 05.2020; SAE-AMS 6470 P 08.2018; SAE-AMS 6472 G 08.2018

SAE-AMS S6709 A SEPT. 2002

CHEMICAL ANALYSIS OF HEAT

U: % HEAT NUMBER: 643615

	C	Mn	Si	P	S	Cr	Ni	Mo	Al
Min.	0.380	0.500	0.200			1.400		0.300	0.950
Max.	0.430	0.700	0.400	0.025	0.025	1.800	0.250	0.400	1.300
Cor.	0.428	0.630	0.221	0.012	0.004	1.610	0.134	0.329	1.071

D.I. ASTM A255 (") : 7.958

JOMINY HARDENABILITY

HARDNESS U. HRC NORM. TEMP.: 1.800-10/10°F

8/16" 12/16"

Min. 50,0 45,0

Max.

Cor. 58,0 58,0

JOMINY Standard (ASTM A255 10 2014); Temperature of (Austenitizing 1.700-10/10°F): Austenitizing 1.700-10/10°F

INCLUSIONS (MICROINCLUSIONS)

Microinclusions std. (ASTM E45 18A 01.06.2018); Type/method (A); A(t): 1,3; A(h): 0,7; B(t): 1,4; B(h): 0,8

C(t): 0; C(h): 0; D(t): 0,7; D(h): 0,5

INCLUSIONS (MACROINCLUSIONS)

Standard (SAE-AMS 2301 L 05.2020); Severity (1): Mean value 0,14; Severity (2): Individual value 0,26

Frequency (1): Mean value 0,15; Frequency (2): Individual value 0,26

MECHANICAL PROPERTIES AS SUPPLIED (CONDITIONS)

Temperature of: (1): Quenching 1.706°F; Time: (1): 1,2 hours; Cooling: (1): AGUA

Temperature media at the start: (1): 86°F; Temperature media at the finish: (1): 91°F

Temperature of: (2): Tempering 1.310°F; Time: (2): 1,2 hours; Cooling: (2): AIRE

Temperature of: (3): Stress Relieved 1.148°F; Time: (3): 6 hours; Cooling: (3): AIRE

Furnaces survey and calibration: OK

MECHANICAL PROPERTIES AS SUPPLIED (TEST)

Specimen Test location: Nucleus

Tensile std. (ASTM A370 -21 01.11.2021); Tensile dir.: Longitudinal

Ts (>=113 KSI): 124 KSI; Ys (1) (Rp (0,2%)) >=90 KSI; Rp (0,2%) 101 KSI; Ys (2) (0,02% KSI): 0,02% 98 KSI; Ys/Ts: 0,81%

E1 (4d >=16%): 4d 22,3%; Z (>=50%): 60,6%

Hardness Standard (ASTM E10 -18 01.07.2018); Hardness at 1/2 radius (241/285 HB): 268 HB; Core hardness (HB): 265 HB

ADDITIONAL TESTS (SUPPLY CONDITIONS)

Decarburization: 0; Grain size std. (ASTM E112 2013 01.10.2013); Grain size: 7

TECHNOLOGY & QUALITY CERTIFIES THAT THE PRODUCT FULL FILLS THE ORDER'S SPECIFICATIONS

APPROVED BY: Manuel Dávila Piris

SIGN:

DATE: 17.05.2023

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REF.: 2001711690000

Analyst of Quality certificates

PO 0001454
 PU 309-175-00-MOODIK-RO

Sidenor

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ISO 9001; IATF 16949; ISO 14001; ISO 45001; ISO 50001; ISO 27001 Y TISAX
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	ROLLED:30.03.2023
MASTER REFERENCE:294845	

Macroetching std.(ASTM E381 17 01.06.2017); Macroetching(1):S 1; Macroetching(2):R 1; Macroetching(3):C 1
 Radioactivity:free <= 0,2pSV/H

NON DESTRUCTIVE TESTS
Surface defects std.(ASTM E709 -15 01.06.2015)
U.T. standard(ASTM A385-A388M -19 01.09.2019); CRACKS CONTROL 100%:OK; ULTRASONIC INSPECTION 100% : O.K.

ADDITIONAL INFORMATION
Reduction ratio:175,69; Initial Product Format:Ingot; Magnetism:<= 10Gauss; MIL-S-5709 Condition "P4
Free of Magnetism; UNS-X24065; acc to SAE-AMS 6472, except pyrometry acc to /SAE-AMS 2750

Material melted and manufactured 100% in Spain through the Electric Arc Furnace (EAF) and Vacuum Degassing route (VOD).

Steel not exposed to Mercury, or to any other metal alloy that is liquid, at ambient temperatures during processing or while in Sidenor's possession.

Material dimensionally controlled, visually inspected, and free from radioactivity (<0,1Bq/g concerning a Co-60).

Steel products were not repaired by welding.
 100% anti mix test performed.

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APPROVED BY:Manuel Dávila Pira

SIGN:

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RRF.:2001711690000

Analyst of Quality certification



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MOR: 416653

SR: 20275312

Packing List / Certificate of Conformance

Sold To: EMI-Schaumburg

Producing Mill: Sideror Aceros Especiales S.L.

Customer PO / Item	Bundle #	Description	Heat # / Lot #	Pieces	Weight Pounds
P257574-423 / 508274	951098	NITR-CFOTS-BRRD-1.5000	643615 / 2259220	67	4.797
				Total	67 - 4.797

PO 001894
PW 309-175-06-MODD1K-RO